Zulfiqar Ahmad Khan

Postdoctoral Fellow (WASP), Department of Computing Science
Umeå University, Umeå, Sweden

zulfiqar.khan@umu.se
+46 76 970 43 40
Google Scholar | ORCID | Umeå University

RESEARCH PROFILE

I am a researcher in AI and deep learning, focusing on computer vision, energy forecasting, edge and resource-efficient AI, and federated learning. I am currently a postdoctoral researcher in the Department of Computing Science at Umeå University, Sweden, through the Wallenberg AI, Autonomous Systems and Software Program (WASP), where I contribute to research projects, co-supervise PhD students, and support research team coordination. During my PhD, I served as Coordinator of the Intelligent Media Laboratory (IMLab), leading a research team of eight members and contributing to research projects, publications, project proposals, and student supervision.

RESEARCH METRICS

49	14	3000+	25	30+	6
Publications	First-author papers	Citations	h-index	i10-index	Korean patents

EDUCATION

Sep 2019 – Feb 2025 **Joint Master's and Ph.D.**, Department of Software, Sejong University, Seoul, South Korea
Thesis: *A Study of Spatiotemporal Feature Analysis using Deep Hybrid Models for Power Forecasting*

Sep 2016 – Aug 2018 **M.Sc. in Computer Science**, Islamia College Peshawar, Pakistan
Thesis: *Towards Smart Home Automation using IoT-enabled Edge-Computing Paradigm* (DIPLab Prototype)

Aug 2014 – Aug 2016 **B.Sc. in Computer Science**, University of Malakand, Dir Lower, Khyber Pakhtunkhwa, Pakistan

RESEARCH EXPERIENCE

May 2025 – present **Postdoctoral Researcher (WASP)**, Department of Computing Science, Umeå University, Sweden
- Researching and implementing open-set recognition methods for computer vision.
- Designing vision transformer architectures with spiking neural networks for energy-efficient visual processing.
- Developing federated learning pipelines for privacy-preserving computer vision on edge devices.
- Developing efficient algorithms for real-time fire detection.
- Co-supervising PhD students and mentoring students in research ideation, algorithm implementation, and manuscript preparation.

Mar 2025 – Apr 2025 **Postdoctoral Researcher**, BrainAI Lab, Kyungpook National University, Daegu, South Korea
- Developed deep learning models for short-term energy forecasting and smart-grid optimization.
- Conducted experiments and evaluated model performance for energy forecasting applications.

Jul 2024 – Mar 2025 **IMLab Coordinator**, Sejong University, Seoul, South Korea
- Led a research team of eight members working on energy informatics, computer vision, and signal processing.
- Co-supervised Master's and PhD students on research planning, algorithm development, experimentation, and manuscript preparation.
- Mentored students in research idea development, algorithm implementation, and experimental evaluation.
- Contributed to multi-year research proposals submitted to the National Research Foundation of Korea (NRF).

Sep 2019 – Jun 2024 **Research Assistant**, Intelligent Media Laboratory, Sejong University, Seoul, South Korea
- Conducted research and development across multiple projects, contributing to algorithm development, experimentation, and publications.
- Collaborated with senior researchers on algorithm implementation and experimental evaluation.
- Implemented research methods and prototypes and evaluated their performance.

Mar 2017 – Aug 2019 **Research Assistant**, Digital Image Processing Laboratory, Islamia College Peshawar, Pakistan
- Supervised bachelor-level research projects in energy informatics, computer vision, and signal processing.

- Prepared research project proposals to support funding applications and laboratory initiatives.
- Participated in research events and workshops to support knowledge exchange.

RESEARCH PROJECTS

Contributed to NRF- and IITP-funded projects in AI, computer vision, energy forecasting, anomaly detection, and materials informatics: algorithm development, project coordination, publications, patents, and project reporting.

May 2025 – present **Wallenberg AI, Autonomous Systems and Software Program (WASP)**
Knut and Alice Wallenberg Foundation

Mar 2025 – Apr 2025 **Brain-Inspired AI Application**
National Research Foundation of Korea (NRF) and Institute of Information and Communications Technology Planning and Evaluation (IITP)

Sep 2024 – Feb 2025 **Materials Synthesis Data Hub**
National Research Foundation of Korea (NRF)

Apr 2024 – Dec 2024 **AI Policy Simulation for Population-Declining Regions**
Institute of Information and Communications Technology Planning and Evaluation (IITP)

Mar 2023 – Feb 2025 **Anomaly Detection for Accident Prevention**
National Research Foundation of Korea (NRF)

Oct 2019 – Dec 2023 **Ensemble Learning and XAI for Energy Analysis**
National Research Foundation of Korea (NRF)

AWARDS & HONORS

Feb 2025 **Outstanding Researcher Award**, Sejong University, Seoul, South Korea

Sep 2023 **Best Paper Award**, Next Generation Computing Conference, Jeju, South Korea

Sep 2019 **Full Scholarship, Joint Master's and Ph.D. Program**, Sejong University, Seoul, South Korea

Aug 2019 **Winner, Business Plan Competition**, ORIC, Islamia College Peshawar, Pakistan

Dec 2019 **Full Scholarship, M.Sc. in Computer Science**, Islamia College Peshawar, Pakistan

RESEARCH RANKINGS & RECOGNITION

2024 – 2025 **Top 2% Scientist**, Updated Science-Wide Author Databases of Standardized Citation Indicators (2024, 2025, 2026)

2021 – 2025 **Scholarly Index Research Rankings**, source

- Fire Detection and Safety Systems: World Rank 14 of 2,458 (top 0.57%); South Korea Rank 3 of 51.
- Energy Load and Power Forecasting: World Rank 170 of 12,721 (top 1.4%); South Korea Rank 6 of 20.
- Solar Radiation and Photovoltaics: World Rank 178 of 8,779 (top 2.1%); South Korea Rank 3 of 21.

PUBLICATIONS

Journal Articles

2026

[J1] AIoT-Enabled Dual-Phase Framework for Industrial Anomaly Detection and Recognition. Adnan Hussain, **Zulfiqar Ahmad Khan**, Hikmat Yar, Sung Wook Baik. *IEEE Internet of Things Journal*.

[J2] A Unified Deep Supervised Network for Effective Animal Voice Recognition. Hikmat Yar, **Zulfiqar Ahmad Khan**, Waseem Ullah, Samee Ullah Khan, Habib Khan, Min Je Kim, Sung Wook Baik. *CMES – Computer Modeling in Engineering & Sciences*.

[J3] Multi-model structure-agnostic framework for enhanced materials discovery in engineering informatics. Altaf Hussain, Muhammad Munsif, **Zulfiqar Ahmad Khan**, Kaleem Ullah, Sung Wook Baik. *Advanced Engineering Informatics*.

[J4] Deep hybrid network with additive attention for accurate population forecasting. Hikmat Yar, Adnan Hussain, **Zulfiqar Ahmad Khan**, Min Je Kim, Sung Wook Baik. *Soft Computing*.

[J5] A novel deep learning framework for battery performance prediction over the operational lifespan. Hikmat Yar, **Zulfiqar Ahmad Khan**, Adnan Hussain, Sang Il Yoon, Seoa Kim, Jungwook Choi, Chan Mi Jeon, Huisu Jeung, Kyungjung Kwon, Sung Wook Baik. *Journal of Energy Storage*.

[J6] Class-incremental learning network for real-time anomaly recognition in surveillance environments. Adnan Hussain, Waseem Ullah, Noman Khan, **Zulfiqar Ahmad Khan**, Hikmat Yar, Sung Wook Baik. *Pattern Recognition*.

2025

[J7] Optimized Cross Module Attention Network and Database for Fire Detection. **Zulfiqar Ahmad Khan**, Fath U Min Ullah, Hikmat Yar, Waseem Ullah, Sung Wook Baik. *Pattern Recognition*.

[J8] EFNet-CSM: EfficientNet with a Modified Attention Mechanism for Effective Fire Detection. Hikmat Yar, Fath U Min Ullah, **Zulfiqar Ahmad Khan**, Min Je Kim, Sung Wook Baik. *Knowledge-Based Systems*.

[J9] Edge-assisted framework for instant anomaly detection and cloud-based anomaly recognition in smart surveillance. Adnan Hussain, Noman Khan, **Zulfiqar Ahmad Khan**, Hikmat Yar, Min Je Kim, Sung Wook Baik. *Engineering Applications of Artificial Intelligence*.

[J10] Dual stream deep attention networks for annual population projection. Adnan Hussain, Hikmat Yar, Noman Khan, **Zulfiqar Ahmad Khan**, Min Je Kim, Sung Wook Baik. *Pattern Analysis and Applications*.

[J11] Hierarchical attention-based framework for enhanced prediction and optimization of organic and inorganic material synthesis. Muhammad Munsif, Altaf Hussain, **Zulfiqar Ahmad Khan**, Min Je Kim, Sung Wook Baik. *Advanced Engineering Informatics*.

[J12] Enhancing real-time fire detection: An effective multi-attention network and a fire benchmark. Taimoor Khan, **Zulfiqar Ahmad Khan**, Chang Choi. *Neural Computing and Applications*.

2024

[J13] DSPM: Dual sequence prediction model for efficient energy management in micro-grid. **Zulfiqar Ahmad Khan**, Shabbir Ahmad Khan, Tanveer Hussain, Sung Wook Baik. *Applied Energy*.

[J14] A modified vision transformer architecture with scratch learning capabilities for effective fire detection. Hikmat Yar, **Zulfiqar Ahmad Khan**, Tanveer Hussain, Sung Wook Baik. *Expert Systems with Applications*.

[J15] TDS-Net: Transformer enhanced dual-stream network for video anomaly detection. Adnan Hussain, Waseem Ullah, Noman Khan, **Zulfiqar Ahmad Khan**, Min Je Kim, Sung Wook Baik. *Expert Systems with Applications*.

[J16] An efficient deep learning architecture for effective fire detection in smart surveillance. Hikmat Yar, **Zulfiqar Ahmad Khan**, Imad Rida, Waseem Ullah, Min Je Kim, Sung Wook Baik. *Image and Vision Computing*.

[J17] Deep multi-scale pyramidal features network for supervised video summarization. Habib Khan, Tanveer Hussain, Samee Ullah Khan, **Zulfiqar Ahmad Khan**, Sung Wook Baik. *Expert Systems with Applications*.

2023

[J18] A Trapezoid Attention Mechanism for Power Generation and Consumption Forecasting. **Zulfiqar Ahmad Khan**, Tanveer Hussain, Waseem Ullah, Sung Wook Baik. *IEEE Transactions on Industrial Informatics*.

[J19] Dual stream network with attention mechanism for photovoltaic power forecasting. **Zulfiqar Ahmad Khan**, Tanveer Hussain, Sung Wook Baik. *Applied Energy*.

[J20] Modelling Electricity Consumption During the COVID-19 Pandemic: Datasets, Models, Results and a Research Agenda. **Zulfiqar Ahmad Khan**, Tanveer Hussain, Amin Ullah, Waseem Ullah, Javier Del Ser, Khan Muhammad, Muhammad Sajjad, Sung Wook Baik. *Energy and Buildings*.

[J21] An Effective Attention-based CNN Model for Fire Detection in Adverse Weather Conditions. Hikmat Yar, Waseem Ullah, **Zulfiqar Ahmad Khan**, Sung Wook Baik. *ISPRS Journal of Photogrammetry and Remote Sensing*.

[J22] A modified YOLOv5 architecture for efficient fire detection in smart cities. Hikmat Yar, **Zulfiqar Ahmad Khan**, Fath U Min Ullah, Waseem Ullah, Sung Wook Baik. *Expert Systems with Applications*.

[J23] Sequential Attention Mechanism for Weakly Supervised Video Anomaly Detection. Waseem Ullah, Fath U Min Ullah, **Zulfiqar Ahmad Khan**, Sung Wook Baik. *Expert Systems with Applications*.

2022

[J24] Randomly initialized CNN with densely connected stacked autoencoder for efficient fire detection. **Zulfiqar Ahmad Khan**, Tanveer Hussain, Fath U Min Ullah, Suneet Kumar Gupta, Mi Young Lee, Sung Wook Baik. *Engineering Applications of Artificial Intelligence*.

[J25] Towards efficient and effective renewable energy prediction via deep learning. **Zulfiqar Ahmad Khan**, Tanveer Hussain, Ijaz Ul Haq, Fath U Min Ullah, Sung Wook Baik. *Energy Reports*.

[J26] Efficient short-term electricity load forecasting for effective energy management. **Zulfiqar Ahmad Khan**, Amin Ullah, Ijaz Ul Haq, Mohamed Hamdy, Gerardo Maria Mauro, Khan Muhammad, Mohammad Hijji, Sung Wook Baik. *Sustainable Energy Technologies and Assessments*.

[J27] Boosting energy harvesting via deep learning-based renewable power generation prediction. **Zulfiqar Ahmad Khan**, Tanveer Hussain, Sung Wook Baik. *Journal of King Saud University – Science*.

[J28] Intelligent dual stream CNN and echo state network for anomaly detection. Waseem Ullah, Tanveer Hussain,

Zulfiqar Ahmad Khan, Umair Haroon, Sung Wook Baik. *Knowledge-Based Systems*.

[J29] A hybrid deep learning-based network for photovoltaic power forecasting. Altaf Hussain, **Zulfiqar Ahmad Khan**, Tanveer Hussain, Fath U Min Ullah, Seungmin Rho, Sung Wook Baik. *Complexity*.

[J30] Optimized dual fire attention network and medium-scale fire classification benchmark. Hikmat Yar, Tanveer Hussain, Mohit Agarwal, **Zulfiqar Ahmad Khan**, Suneet Kumar Gupta, Sung Wook Baik. *IEEE Transactions on Image Processing*.

[J31] WEENet: an intelligent system for diagnosing COVID-19 and lung cancer in IoMT environments. Khan Muhammad, Hayat Ullah, **Zulfiqar Ahmad Khan**, Abdul Khader Jilani Saudagar, Abdullah AlTameem, Mohammed AlKhathami, Muhammad Badruddin Khan, Mozaherul Hoque Abul Hasanat, Khalid Mahmood Malik, Mohammad Hijji, Muhammad Sajjad. *Frontiers in Oncology*.

2021

[J32] Atrous convolutions and residual GRU based architecture for matching power demand with supply. Samee Ullah Khan, Ijaz Ul Haq, **Zulfiqar Ahmad Khan**, Noman Khan, Mi Young Lee, Sung Wook Baik. *Sensors (MDPI)*.

[J33] Towards smart home automation using IoT-enabled edge-computing paradigm. Hikmat Yar, Ali Shariq Imran, **Zulfiqar Ahmad Khan**, Muhammad Sajjad, Zenun Kastrati. *Sensors (MDPI)*.

[J34] An efficient anomaly recognition framework using an attention residual LSTM in surveillance videos. Waseem Ullah, Amin Ullah, Tanveer Hussain, **Zulfiqar Ahmad Khan**, Sung Wook Baik. *Sensors (MDPI)*.

2020

[J35] Electrical energy prediction in residential buildings for short-term horizons using hybrid deep learning strategy. **Zulfiqar Ahmad Khan**, Amin Ullah, Waseem Ullah, Seungmin Rho, Mi Young Lee, Sung Wook Baik. *Applied Sciences*.

[J36] Towards efficient electricity forecasting in residential and commercial buildings: A novel hybrid CNN with a LSTM-AE based framework. **Zulfiqar Ahmad Khan**, Tanveer Hussain, Amin Ullah, Seungmin Rho, Mi Young Lee, Sung Wook Baik. *Sensors (MDPI)*.

[J37] A novel CNN-GRU-based hybrid approach for short-term residential load forecasting. Muhammad Sajjad, **Zulfiqar Ahmad Khan**, Amin Ullah, Tanveer Hussain, Waseem Ullah, Mi Young Lee, Sung Wook Baik. *IEEE Access*.

Conference Papers

[C1] GeoSwin: Hierarchical Representation Learning and a Large-Scale Earth Observation Benchmark. Muhammad Fayaz, **Zulfiqar Ahmad Khan**, Lien Minh Dang, Hyeonjoon Moon. *CVPR 2026 Workshop* (accepted).

[C2] TinySEED: A Lightweight Transformer Architecture with Channel-Attention for DDoS Detection. Yinuo Zhang, **Zulfiqar Ahmad Khan**, Monowar Bhuyan. *ACM SAC 2026* (accepted).

[C3] Catastrophic Forgetting Resilient One-Shot Incremental Federated Learning. Obaidullah Zaland, **Zulfiqar Ahmad Khan**, Monowar Bhuyan. *ICBDA 2026* (accepted).

[C4] Dual Deep Learning Network for Abnormal Action Detection. Fath U Min Ullah, **Zulfiqar Ahmad Khan**, Sung Wook Baik, Estefania Talavera, Saeed Anwar, Khan Muhammad. *IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*, 2024.

[C5] Dataset Standardization for Effective Solar Power Forecasting: A Comprehensive Analysis. **Zulfiqar Ahmad Khan**, Waseem Ullah, Hikmat Yar, Noman Khan, Min Je Kim, Sung Wook Baik. *9th International Conference on Next Generation Computing (ICNGC)*, Danang, Vietnam, 2023.

[C6] Comparative Analysis of Solar Power Generation Forecasting Models for Identical Latitude Countries Data. Noman Khan, Waseem Ullah, **Zulfiqar Ahmad Khan**, Adnan Hussain, Min Je Kim, Sang Il Yoon, Sung Wook Baik. *9th International Conference on Next Generation Computing (ICNGC)*, Danang, Vietnam, 2023.

[C7] Solar Power Prediction using Dual Stream CNN-LSTM Architecture. **Zulfiqar Ahmad Khan**, Noman Khan, Su Min Lee, Sang Il Yoon, Mi Young Lee, Sung Wook Baik. *Korea Next Generation Computing Society Spring Conference*, 2022.

[C8] Efficient vehicle detection in aerial scenes. Habib Khan, **Zulfiqar Ahmad Khan**, Waseem Ullah, Min Je Kim, Mi Young Lee, Sung Wook Baik. *Korea Next Generation Computing Society Spring Conference*, 2022.

[C9] PV-ANet: Attention-Based Network for Short-term Photovoltaic Power Forecasting. Muhammad Munsif, Habib Khan, **Zulfiqar Ahmad Khan**, Altaf Hussain, Fath U Min Ullah, Mi Young Lee, Sung Wook Baik. *8th International Conference on Next Generation Computing*, 2022.

[C10] Disasters Scenes Classification Based on Unmanned Aerial Vehicles Using Lightweight CNN. Altaf Hussain, Samee Ullah Khan, Fath U Min Ullah, **Zulfiqar Ahmad Khan**, Mi Young Lee, Sung Wook Baik. *7th International Conference on Next Generation Computing*, 2021.

[C11] Fire detection through effective vision transformers. Hikmat Yar, Tanveer Hussain, **Zulfiqar Ahmad Khan**, Mi Young Lee, Sung Wook Baik. *Korean Society for Next Generation Computing Journal*, 17(5):21–30, 2021.

[C12] An Adaptive Filtering Technique for Segmentation of Tuberculosis in Microscopic Images. **Zulfiqar Ahmad Khan**, Waseem Ullah, Amin Ullah, Seungmin Rho, Mi Young Lee, Sung Wook Baik. *4th International Conference on Natural Language Processing and Information Retrieval (NLPIR)*, 2020.

Manuscripts Under Review / Submitted

[S1] FedSHIELD: Adversarial Honeypot-Guided Client Screening Framework for Robust Federated Learning. **Zulfiqar Ahmad Khan**, Obaidullah Zaland, Monowar Bhuyan. *Submitted to ICLR 2027*.

[S2] Fidelity is Not Enough: Tabular Anomaly Detection under Synthetic Distribution Shift. Yinuo Zhang, **Zulfiqar Ahmad Khan**, Monowar Bhuyan. *Submitted to ICLR 2027*.

[S3] SeaFormer: Single-Pass Uncertainty for Energy-Efficient Underwater. **Zulfiqar Ahmad Khan**, Obaidullah Zaland, Monowar Bhuyan. *Submitted to ICRA 2027*.

[S4] Reinforcing Diversity: RL-Guided Experts for Open-Set Recognition. **Zulfiqar Ahmad Khan**, Fath U Min Ullah, Monowar Bhuyan. *Submitted to AAAI 2027*.

[S5] Attention Transfer for One-Shot Federated Learning. Obaidullah Zaland, V. Subramaniam, **Zulfiqar Ahmad Khan**, V. Subbaraju, Monowar Bhuyan. *Submitted to WACV 2027*.

[S6] Privacy Risk Aware Federated Split Learning for Visual Grounding. Obaidullah Zaland, V. Subramaniam, **Zulfiqar Ahmad Khan**, V. Subbaraju, Monowar Bhuyan. *Under review*.

[S7] Towards Better Generalization in Federated Learning with Attention Regularization. Obaidullah Zaland, V. Subramaniam, **Zulfiqar Ahmad Khan**, V. Subbaraju, Monowar Bhuyan. *Under review*.

[S8] FLARE-Net: A Federated Spiking Vision Transformer for Resource-Constrained Fire Detection. **Zulfiqar Ahmad Khan**, Monowar Bhuyan. *Submitted to IEEE Transactions on Image Processing*.

[S9] SFFormer: Saliency Aware Gated Factorized Attention for Spiking Neural Networks. **Zulfiqar Ahmad Khan**, Obaidullah Zaland, Monowar Bhuyan. *Submitted to Neural Networks*.

[S10] Incremental Learning Enhanced Dual-Stream Deep Attention Network for Power Forecasting. **Zulfiqar Ahmad Khan**, Sung Wook Baik. *Applied Energy* (major revision submitted).

[S11] GeoSpikeFormer: A Neuromorphic Transformer and Comprehensive Benchmark for Large-Scale Land Cover Mapping. Muhammad Fayaz, **Zulfiqar Ahmad Khan**, Hyeonjoon Moon. *Submitted to IEEE Transactions on Image Processing*.

[S12] Attention Enhanced YOLOv8 Architecture and a Benchmark for Fire Detection in Smart Surveillance and Remote Sensing. Hikmat Yar, **Zulfiqar Ahmad Khan**, Waseem Ullah, Min Je Kim, Sung Wook Baik. *Submitted to Remote Sensing of Environment*.

[S13] Ensemble Feature Learning with Spatiotemporal Share Attention Memory for Anomaly Detection. Waseem Ullah, Fath U Min Ullah, **Zulfiqar Ahmad Khan**, Min Je Kim, Sung Wook Baik. *Submitted to IEEE Transactions on Cybernetics*.

[S14] A Dual Stream Attention Network for Real-World Anomalies Recognition. Waseem Ullah, **Zulfiqar Ahmad Khan**, Hikmat Yar, Sung Wook Baik. *Submitted to IEEE Transactions on Circuits and Systems for Video Technology*.

[S15] Anomaly Recognition and Localization in Surveillance Systems. Waseem Ullah, **Zulfiqar Ahmad Khan**, Hikmat Yar, Sung Wook Baik. *Submitted to Engineering Applications of Artificial Intelligence*.

[S16] Deep Learning-Enhanced Driver Behavior Assessment with Spatial Awareness for Safe Transportation. Hikmat Yar, **Zulfiqar Ahmad Khan**, Waseem Ullah, Min Je Kim, Sung Wook Baik. *Submitted to Advanced Engineering Informatics*.

PATENTS (KOREA)

[P1] Power consumption prediction system of residential and commercial building using hybrid convolution neural network and method thereof. Sung-Wook Baik, Seung-Min Rho, Mi-Young Lee, **Zulfiqar Ahmad Khan**, Tanveer Hussain, Amin Ullah. Reg. No. 1025005480000.

[P2] Anomaly recognition method and system based on LSTM. Sung-Wook Baik, Mi-Young Lee, Waseem Ullah, Tanveer Hussain, **Zulfiqar Ahmad Khan**. Reg. No. 1026012330000.

[P3] Method for predicting renewable energy consumption and production based on deep learning and apparatus thereof. Sung-Wook Baik, Mi-Young Lee, Altaf Hussain, **Zulfiqar Ahmad Khan**, Fath U Min Ullah, Minje Kim, Sumin Lee. Reg. No. 1025335080000.

[P4] Method for predicting energy production and consumption data using double sequence deep learning model, and apparatus thereof. Sung-Wook Baik, Mi-Young Lee, **Zulfiqar Ahmad Khan**, Minje Kim, Sangil Yoon, Sumin Lee. Reg. No. 1025218070000.

[P5] Method for predicting energy consumption and production based on hybrid convolutional neural network and regression neural network and apparatus thereof. Sung-Wook Baik, Mi-Young Lee, **Zulfiqar Ahmad**

Khan, Fath U Min Ullah, Minje Kim, Sangil Yoon, Sumin Lee. Reg. No. 1025218080000.

[P6] Method for predicting renewable energy consumption and production based on deep learning and apparatus thereof. Sung-Wook Baik, Mi-Young Lee, **Zulfiqar Ahmad Khan**, Tanveer Hussain, Minje Kim. Appl. No. 1020220061428 (submitted).

TECHNICAL SKILLS

Programming	Python, C++, PHP
Deep learning	PyTorch, TensorFlow, Keras
Computer vision	OpenCV, scikit-image, Pillow
Data analysis	Pandas, NumPy, Matplotlib, Seaborn; time-series analysis, regression models
ML methods	Supervised and unsupervised learning, active learning, incremental learning, federated learning
IoT & databases	MQTT, Raspberry Pi, Arduino; MySQL, SQLite, XAMPP
Tools	Git, GitHub, Jupyter, Spyder, MATLAB Simulink, LaTeX, Microsoft Office

ACADEMIC SERVICE

Peer review	60+ reviews (record on ORCID) for IEEE Transactions on Cybernetics; IEEE Transactions on Industrial Informatics; Engineering Applications of Artificial Intelligence; Knowledge-Based Systems; Information Processing and Management; Artificial Intelligence Review; Journal of King Saud University – Computer and Information Sciences; Cluster Computing; Energy Efficiency; Stochastic Environmental Research and Risk Assessment; Scientific Reports; IET Generation, Transmission & Distribution; IET Renewable Energy.

REFERENCES

Prof. Sung Wook Baik
Ph.D. Supervisor
Professor and Director, IMLab
Department of Software
Sejong University, Seoul, Korea
sbaik@sejong.ac.kr
+82 10 2439 9436

Dr. Monowar Bhuyan
Postdoc Advisor
Assistant Professor
Dept. of Computing Science
Umeå University, Sweden
monowar.bhuyan@umu.se
+46 90 786 67 05

Dr. Amin Ullah
Former Lab Coordinator
Senior AI/ML Software Engineer
Boeing Research & Technology
Seattle, WA, USA
qamin3797@gmail.com
+1 541 908 7580